TOPIC 3: ELECTRIC CURRENT AND DIRECT CURRENT CIRCUITS

1. A 6.0 Ω resistor is placed across the terminal of a 12.0 V battery. Calculate the

   (a) current flows through the resistor.
   (b) magnitude of charge passes through the resistor in 2.0 s.

   [2 A, 4 C]

2. A current of 5.0 A flows in a copper wire for 5 s. Calculate the

   (a) number of electron that flows in the wire within that time.
   (b) amount of charge flows in the wire for 10 s.

   [1.56×10^{20}, 50 C]

3. The aluminium wire used in the high voltage power lines has a cross sectional area of 5.0×10^{-4} m^2. What is the resistance of 15 km of this wire?
   (Given $\rho_{\text{aluminium}} = 2.82 \times 10^{-8}$ Ω m)

   [0.846 Ω]

4. A cylindrical copper cable across a potential difference of 1.5 V carries a current of 1200 A. If the length of the cable is 0.32 m, what is the radius of the cable?
   (Given $\rho_{\text{copper}} = 1.72 \times 10^{-8}$ Ω m)

   [1.18×10^{-3} m]

5. A carbon wire has a resistance of 1 Ω and a diameter of 1 mm. If the resistivity of carbon is 3.5×10^{-5} Ω m, what is the length of the wire?

   [0.022 m]

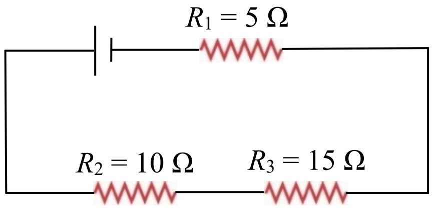
6. 

   FIGURE 1

   Three resistors, R_1, R_2 and R_3 are connected to a battery as shown in **FIGURE 1**. Calculate the effective resistance of the circuit.

   [30 Ω]

7.

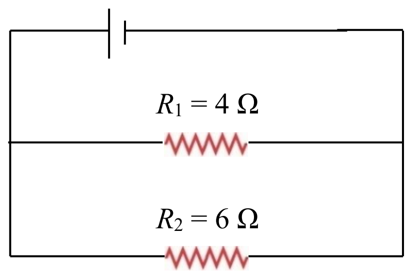


FIGURE 2

Two resistors, R_1 and R_2 are connected to a battery as shown in **FIGURE 2**. Calculate the effective resistance of the circuit.

[2.4 Ω]

8.

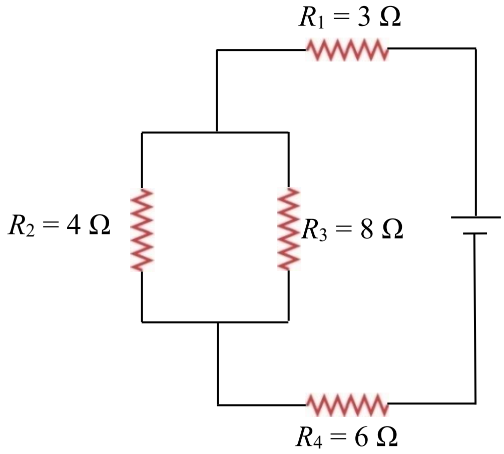


FIGURE 3

Based on **FIGURE 3**, calculate the effective resistance of the circuit.

[11.67 Ω]

9.

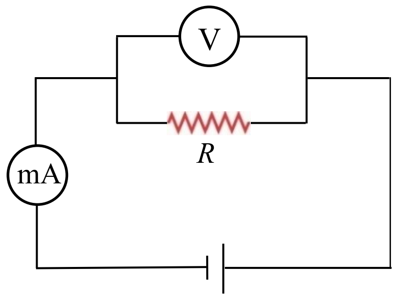


FIGURE 4

FIGURE 4 shows the electric circuit is used to determine the resistance of voltmeter. When the resistance R is 100 Ω, the reading of milliammeter is 30 mA and the reading of voltmeter is 2.7 V. Calculate the resistance of the voltmeter.

[900 Ω]

10.

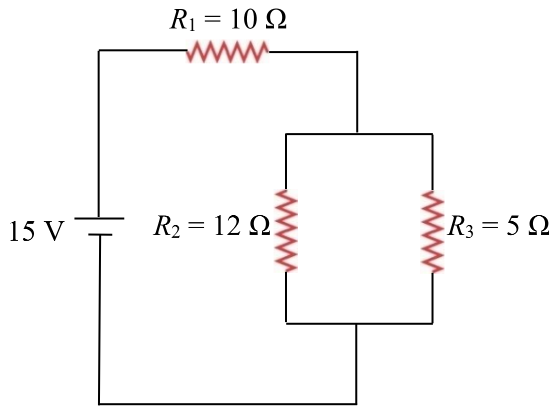


FIGURE 5

Three resistors R_1, R_2 and R_3 are connected to a power supply of 15 V as shown in **FIGURE 5**. If the current that flows in R_1 is 1.11 A, calculate the

(a) potential difference across resistor R_1.
(b) potential difference across resistor R_3.

[11.1 V, 3.9 V]